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Chapter 1

Introduction

1.1 Background of the Study

The landscape of retail is basically unrecognizable, thanks to e-commerce platforms steamrolling old-school shopping. Now, when customers are eyeing complex gear like laptops, it's not in-store demos that sway them—it's the avalanche of user reviews online. These reviews have genuinely become the new "word-of-mouth," bridging the gap created when you can't actually poke at the product before dropping your cash.

Now, the laptop market? A total minefield for decision-making. Products sprawl across specs—processors, RAM, battery juice, screen fidelity, portability, aesthetics, you name it. Each factor gets dissected by different user groups, leaving a breadcrumb trail of critiques that, in aggregate, represent a surplus of granular, user-driven "test data."

But here's the kicker: the sheer *amount* of feedback is both a blessing and a nightmare. Sure, there's a wealth of data, but what regular user has hours to trawl through a few thousand reviews just to decide whether a hinge design is trash or top-tier? It's information overload, plain and simple—equally problematic for end-users and manufacturers itching for actionable feedback.

Most legacy analysis methods? Honestly, not up to the task. Number crunching star ratings? That's just surface-level; it totally fails to reflect the duality you see in long-form reviews (e.g., "Great performance, but the fan sounds like a jet engine"). You're leaving useful specifics on the table with that five-star tally.

So, sentiment analysis shows up—think of it as Natural Language Processing for product reviews. These algorithms scan reviews, parse out attitudes, and assign sentiment tags, automating what would otherwise be weeks of manual effort. Classical ML models like Naive Bayes or SVM get you basic polarity—yeah, positive or negative—but they're barely scratching the surface. Advanced architectures, like BERT, seriously up the game. They parse nuance, resolve context, and don't get tripped up by tricky syntax or compound feedback buried in a review.

One huge complication: data is skewed. Positive reviews flood in, negatives are rarer. Robust models need to recognize insight in skewed datasets or they'll yield totally warped results. Assessment under both balanced and imbalanced data conditions is non-negotiable for real-world deployment.

This study steps up by systematically benchmarking an armada of sentiment analysis models—classic and cutting-edge—against a beastly dataset: 24,113 laptop reviews from major e-commerce players. The goal? Not just academic novelty, but clear, evidence-driven recommendations for the e-commerce industry. The idea is to pull genuinely actionable intelligence from review data—useful for both researchers hungry for new benchmarks and businesses wanting to tighten their feedback loop with the actual market.

1.2 Importance of Sentiment Analysis in Indian E-Commerce

Sentiment analysis has a transformative role to play in the emerging Indian e-business landscape as it provides much richer information beyond the traditional customer satisfaction measures. Indian customers provide millions of product reviews every year through e-shopping sites from Flipkart to Amazon, and capturing such sentiments has become an important element of business competitiveness and customer-focused innovation.

1. Informed Decision-Making for Buyers

Shoppers in India are increasingly looking to peer ratings and reviews before buying - more than 95% look at reviews before making a purchase and 77% believe they are a significant trust factor. Sentiment analysis analyzes an incredible volume of unstructured comments, provides concise summaries, and uncovers core positive and negative themes within those comments. This allows prospective buyers to make informed, confident buying decisions based on the community's insight, not advertisements' rhetoric.

2. Product and Service Improvement Using Data

Brands and sellers use sentiment analysis to identify recurring problems or unsatisfied needs such as quality problems, delivery issues, or desired features. For example, if a series of reviews mention a poor battery or size problem, manufacturers can make addressing such with future models or updating product information a priority. Such a feedback system assists with rapid product innovation and return rates and dissatisfaction.

3. Better Customer Service and Retention

Artificial intelligence and machine learning algorithms can automatically detect negative or high priority reviews and allow customer service teams to respond right away and resolve issues and turn unhappy customers into ambassadors of the brand. Sentiment analysis

implementations at business organizations have witnessed up to a 15% increase in customer retention rate through proactive customer contact.

4. Competitive Benchmarking and Brand Reputation

With real-time sentiment monitoring, e-commerce and brands can evaluate their performance against competitors, track reputation trends, and distinguish strengths to promote from weaknesses to mitigate. Sentiment analysis can also be conducted at the aspect or attribute level, or breakdown opinions at the feature or attribute level and allows for finely grained competitive insight—a requirement in India's price-sensitive and value-conscious market.

5. Meeting the Indian Consumer Expression Diversity

India's use of a code-mixed language (e.g. Hindi-English mixes), and its multilingualism, makes manual review analysis difficult and inefficient. Advanced sentiment analysis software trained with cultural literacy of the Indian language context are more readily able to interpret local expressions, and local idioms, in addition to cultural shades of meaning, and provide businesses more accurate and useful conclusions.

6. Strategic Marketing and Recommendation Systems Campaign messaging is based on sentiment trends, and refer-a-friend initiatives focus first on customers with some measure of happiness; and adaptive recommendation engines will consider sentiment at the time of rendering the recommendation considering product credibility. From sentiment indicators--especially the purchasing-validated ones, sites can separate true from false and point new customers to credible products. The sentiment analysis overall allows for e-commerce in India to unlock valuable customer voice at scale and provide new business opportunities, all while enabling a degree of transparency and trust in a shopping ecosystem, which enables higher buyer confidence, less likelihood of dissatisfaction, and operational flexibility for brands and merchants.

Chapter 2

Review of Literature

1. Alzahrani et al. (2022) - "Development of an Intelligent System with Deep Learning Algorithms Supporting Sentiment Analysis of E-Commerce Product Reviews"

Purpose: A deep learning intelligent sentiment classification system designed to categorizing falsely based e-commerce product (non)reviews was developed and tested.

Method: The researchers used the CNN-LSTM and LSTM models on datasets consisting of Amazon product reviews covering reviews from cameras, laptops, mobile phones, tablets, televisions, and video surveillance (13,057 reviews total). Sentiment scoring using lexicon was applied. The researchers preprocessed and performed advanced data preprocessing including converting all characters to lowercase, removal of stopwords and then tokenized the reviews.

Key Results: The CNN-LSTM hybrid model achieved an accuracy of 94% as compared to the individual LSTM with an accuracy of 91%. The hybrid system also classified better true positive (TP) (TP of 83.54% as compared to TP of 82.24%) and false positive (FP) measures (FP of 5.28% as compared to FP of 6.39%). The cross-validation results confirm that the suggested model was stable with the different product groups.

Conclusion: This study supports the effectiveness of using CNN and LSTM architecture for e-commerce sentiment mining, which highlights the importance of identification falsely based reviews and non-reviews from web sites.

2. Bellar et al. (2024) - "Sentiment Analysis Predicting Product Reviews for E-Commerce Recommendations Using Deep Learning and Transformers"

Aim: The goal is to compare and benchmark diverse deep learning models, specifically methods based on transformers, for sentiment analysis involving e-commerce recommendation systems against basic machine learning methods and today's deep learning models.

Approach: The researchers conducted rigorous experiments using the Women's Clothing E-Commerce Reviews dataset (22,641 reviews) and many methods: CNN, RNN, Bi-LSTM with the Word2Vec and FastText embedding methods, a variety of BERT models (BERT, RoBERTa, ALBERT), and ensemble methods for both 3-class and 5-class sentiment classification tasks.

Key Insights: The best transformer result was achieved by RoBERTa, with an overall accuracy of 87.69% at the 5-class configuration. Word2Vec with RNN achieved the best neural network result (94.85% accuracy at the 5-class configuration). The ensemble models achieved the best configuration by a significant margin, and the best result overall (accuracy of 96.2%) with CNN-RNN-Bi-LSTM. Machine learning configurations performed significantly worse (accuracy of 55-66%).

Contributions: Our work provides vital benchmarking evidence towards choice of a methodology in e-commerce sentiment analysis with demonstration of the superiority of deep learning methodologies over baseline and a consideration towards using hybrid methodologies for sentiment analysis.

3. Huang et al. (2023)-"Sentiment Analysis in E-Commerce Platforms: A Review of Current Techniques and Future Directions"

Purpose: This systematic research article reviewed recent sentiment analysis methods incorporated in e-commerce websites and identified patterns, popular websites and implications for future research, based on 54 articles published between 2018-2022.

Method: The researchers conducted a thorough review with definitive keywords (for help across various databases including IEEE, Science Direct, Google Scholar, with strict inclusion and exclusion criteria in order to converge the experimental articles with the clear use of sentiment analysis and dealing with e-commerce problems.

Results: The review identified that 48% of the research applied machine learning methods and 44% used deep learning methods; Amazon was the most popular source (17 of the 50 articles), followed by social media. Traditional methods used frequently were SVM and Naive Bayes and the frequently used deep learning methods were LSTM, CNN, and BERT.

Significance: It determined major research directions such as the requirement of universal models that are applicable across languages and domains, aspect-based sentiment analysis, implicit aspect detection, sarcasm recognition, and fine-grained sentiment analysis.

4. Yang et al. (2020) - "Sentiment Analysis for E-Commerce Product Reviews in Chinese Based on Sentiment Lexicon and Deep Learning"

Objective: We proposed the SLCABG model and combined sentiment lexicon with CNN and attention-driven Bidirectional GRU to explore the sentiment of Chinese e-commerce product reviews.

Methodology: Sentiment lexicon weighting with BERT word embedding and CNN were used first for local feature extraction and BiGRU with attention mechanism were used for context modeling. Corpus consisted of 100,000 book reviews from Dangdang.com and both 10-fold and 5×2 cross-validation were used.

Key Findings: Our SLCABG model outperformed existing approaches (NB, SVM) and single deep learning models (CNN, BiGRU).

The optimal configuration was using 35,000 dictionary words and maximum sentence length input trained for 8 iterations with and dropout of 0.4. Model performance improved tremendously when applying sentiment lexicon weights versus unweighted word vectors.

Significance: Our study advocates the utility of mixed lexicon-based and deep learning approaches, as well as future use toward exploring their approaches in Chinese e-commerce.

5. Munna et al. (2020) - "Sentiment Analysis and Product Review Classification in E-Commerce Platform"

Objective: The study was created to develop sentiment analysis methods of Bangla texts that can help address multilingual sentiment analysis while working in a context of a regional language settings for e-commerce websites.

Methodology: The researchers employed fairly typical machine learning techniques for Bangla product review classification while dealing with the associated concerns of processing non-English text such as proper tokenization, extraction of features and treatment of features in a regional language.

Key Findings: Sentiment classification and product review classification were achieved at an accuracy of 84% and 69% respectively on Bangla texts. The report recorded

instances of issues associated with processing regional language Suffixed language texts such as mixed language reviews and cultural contextualization.

Significance: The study contributes to the important emerging field of multilingual Sentiment Analysis, and takes a multi-lingual Sentiment Analysis approach to a local languages in an e-commerce context.

6. "Application of Sentiment Analysis in E-Commerce"

Objective: How traditional lexicon-based and machine learning sentiment analysis can be applied to an Amazon product review dataset for improving support of customer satisfaction and recommender system.

Methods: The authors compared lexicon-based approaches (using SentiWordNet) with supervised machine learning classifiers (Naive Bayes, Support Vector Machine, and Random Forest) on a set of 10,000 Amazon product reviews for electronics and apparel products. Preprocessing consisted of tokenization, stemming, and stop word removal. Performance evaluation consisted of accuracy, precision, recall, and F1-score.

Key Findings: Lexicon-based approaches yielded an accuracy of 71% but were limited by context-dependent sentiment based on lexicon. Supervised machine learning classifiers using Support Vector Machine yielded an accuracy of 85% and an F1-score of 0.84, yielding more robust classification accuracy across varied vocabulary compared to the comparable supervised classifiers.

Significance: This research highlights practical tradeoffs between lexicon-based simplicity versus machine learning robustness and suggests feature hybridization for e-commerce sentiment analysis tasks.

7. "Transforming Sentiment Analysis for E-Commerce Product and Service Reviews"

Objective: Advanced text-mining methods, such as word embeddings and transformer models, are reviewed for sentiment analysis of service quality in e-commerce applications.

Methodology: The authors compared several embedding approaches (Word2Vec, GloVe, FastText) and different classifiers (BiLSTM, CNN, BERT-based) on a hotel and electronics service review dataset (15,000 reviews). They tested on overall sentiment classification and aspect-based scores (responsiveness, reliability).

Key Findings: FastText embeddings also performed marginally better than Word2Vec (FastText+BERT classifier scored 92%). BERT also surpassed BiLSTM by 5% on aspect-based sentiment extraction, especially for subtle service features. The integration of embeddings with transformer fine-tuning decreased training time by 20% without compromising accuracy.

Significance: The paper emphasizes the efficiency and accuracy of transformer models for both overall and aspect-level sentiment analysis, suggesting BERT fine-tuning as the best practice for service-oriented e-commerce reviews.

8. "Exploring E-Commerce Product Experience Based on Fusion Sentiment Analysis Method"

Objective: This paper introduces a fusion sentiment analysis approach by integrating lexicon-based, statistical, and deep learning approaches to examine user experience on e-commerce sites.

Methodology: The authors created a three-stage pipeline: (1) lexicon scoring with an extended opinion lexicon; (2) statistical feature extraction (TF-IDF, review length, punctuation count); (3) deep learning classification with BiLSTM. The pipeline was tested on 12,500 user reviews from a Chinese electronics platform.

Key Findings: The fusion method reached 89% overall accuracy, outperforming individual BiLSTM (85%) and lexicon-only (73%) methods. Feature fusion enhanced aspect-based detection of design and performance sentiments by 8%. Early fusion (merging together features before classification) outperformed late fusion (ensemble voting) by 4%.

Significance: Demonstrates that integrating multiple sentiment signals yields more robust and nuanced insights into product experience, validating the efficacy of feature-level fusion for e-commerce review analysis.

9. “Measuring E-Commerce Service Quality”

Objective: Despite not being a pure sentiment analysis research, this paper constructs an E-S-QUAL model to quantify service quality factors—efficiency, fulfillment, system

availability, and privacy—based on customer survey responses and review text content analysis.

Methodology: The researchers gathered 8,000 user reviews and survey responses related to Amazon services. They conducted sentiment analysis on review content via SVM and confirmatory factor analysis of survey measures. Correlation analysis associated sentiment scores with dimensions of perceived service quality.

Key Findings: Fulfillment and responsiveness sentiment scores were highly correlated (r = 0.72) with satisfaction measured via surveys. Neutral and negative sentiments within system availability reviews explained 84% of self-reported service outages. Privacy issues had less strong textual sentiment cues, indicating implicit variables in user attitude.

Significance: Supports sentiment analysis as a surrogate measure for classical service-quality measurement, providing an automatic method for tracking e-commerce platform performance.

10. "Sentiment Analysis of Product Reviews for E-Commerce Recommendation Based on Machine Learning"

Objective: This research assesses the effect of using sentiment features within product recommendation systems on e-commerce websites[97f62a29].

Methodology: The authors mined sentiment polarity scores via SVM and Naive Bayes from the 9,200 user reviews as well as from the users purchase history of an electronics store and integrated them into collaborative filtering recommendation model.

Key Findings: Adding sentiment enhanced recommendation precision by 12% and recall by 9%. SVM-derived sentiment features performed better than Naive Bayes features by 4% in terms of recommendation accuracy. Hybrid recommendation (sentiment weighting + collaborative filtering) boosted total F1-score from 0.68 to 0.77.

Significance: Points out the importance of sentiment-aware recommendations, showing how review analysis can enhance personalized product recommendations and increase user engagement.

Chapter 3

Research Methodology / Design

The research adopts a quantitative experimental design to evaluate sentiment analysis techniques on 24,113 Flipkart laptop reviews spanning six major brands (Apple, HP, Lenovo, ASUS, Acer, MSI) from January 2023 to July 2025. Reviews were collected via Flipkart's public API and a validated Kaggle dataset, excluding non-English or empty texts. Preprocessing in Python involved cleaning HTML, URLs, and special characters; normalizing emojis; expanding contractions; lowercasing; tokenizing with spaCy; removing stopwords; and lemmatizing. Reviews were labeled negative (1–2 stars), neutral (3 stars), or positive (4–5 stars).

Given the original class imbalance (81.3% positive, 10.3% negative, 8.4% neutral), two balanced variants were created: an undersampled dataset reducing the positive class to 5,000 samples, and a SMOTE-oversampled dataset bringing all classes to 5,000 samples. Feature extraction for traditional models employed TF-IDF vectorization (top 10,000 unigrams and bigrams) plus review metadata (length, punctuation count). Deep learning models used pretrained Word2Vec and FastText embeddings (300 dimensions) and BERT-base multilingual embeddings (768 dimensions).

Four classifiers were implemented: Multinomial Naive Bayes, L2-regularized Logistic Regression, linear-kernel Support Vector Machine, and a BERT-base fine-tuned model (three epochs, learning rate 2e-5, batch size 16). Models were trained on 80% of data, validated on 10%, and tested on 10%, with stratified splits. Aspect-based analysis extracted sentences containing feature terms (e.g., "battery life") via dependency parsing and classified them using the BERT model. Performance metrics included accuracy, macro-averaged precision, recall, F1-score, confusion matrices, and ROC-AUC for binary sentiment tasks. The methodology ensures a rigorous, reproducible comparison of sentiment analysis approaches on Indian e-commerce laptop reviews.

3.1 Objectives of Research

1. To compare advanced sentiment analysis model with traditional models for accurately classifying e-commerce customer reviews.
2. Comparison of Models Performance on unbalanced and balanced dataset

3. To implement aspect-based sentiment analysis for extracting sentiments related to specific product/service features.
4. To analyze the correlation between numerical ratings and textual sentiment expressed in reviews, identifying any inconsistencies or patterns.

3.2 Statement of the Problem

In India's fast-growing online shopping space, laptops are a complicated item. Customers really depend on what other buyers say to help them pick one. Star ratings are fast, but they don't show all the small details about things like battery length, how it runs, looks, and how much it costs. It's too hard to read through thousands of written reviews for each type, so both shoppers and stores miss out. Regular feeling-reading tools often miss the smaller points and have issues with uneven info, unclear words, and mixed-up language that's normal in Indian reviews. As a result, online stores and companies don't have a way to easily take out good feedback from buyer reviews. This research looks at these problems via making and testing common machine learning and newer feeling-reading models. It checks how they do with even and uneven info, and uses aspect-based sentiment extraction to find user sentiments and feelings.

3.3 Need for the Study

Okay, so India's e-commerce market is blowing up, and that means tons of customer reviews, especially for laptops on places like Flipkart. The problem? All that feedback is just a giant wall of text. It's tough for companies and shoppers to get anything useful out of it.

Right now, people mostly look at star ratings, but that misses the small stuff, like what people really think about the battery or screen. Plus, reading through all those reviews by hand takes forever and people make mistakes, especially 'cause Indian reviews can be kinda confusing, with slang, mixed languages, and sarcasm all over the place.

Sentiment analysis could fix this by automatically figuring out how people feel about products from their reviews. But a lot of studies only look at general good or bad feelings. What we really need is to know what they feel on different parts. Also, most reviews are positive, which

throws off the machine learning and makes critical negative feedback hard to find. Fancy models like BERT could be more exact, but we don't know how it compares with simpler methods for Indian e-commerce.

So, this study will test out different methods, both simple and fancy, on good and bad sets of reviews. It'll also pull out how folks feel about different aspects of the product and will match these feelings, from the text, with the number ratings they gave. This should give everyone in the e-commerce area solid way to use what customers are saying to improve products, make people happier, and make smarter moves.

3.4 Scope for the Study

This research looks at how people feel about laptops, using customer reviews written in English. The reviews came from Flipkart and a well-known Kaggle dataset. We gathered opinions on six popular laptop brands: Apple, HP, Lenovo, ASUS, Acer, and MSI. The data includes reviews posted from January 2023 to July 2025.

The study sorts opinions into three groups: positive, neutral, and negative. These groups are based on the star ratings customers gave. We used both simpler machine learning methods, like Naive Bayes, Logistic Regression, and SVM, and a more complex method using a BERT-based transformer model.

A key challenge was the uneven number of reviews in each sentiment category (class imbalance). To handle this, we tested the models on three versions of the dataset: the original, one with fewer of the most common reviews (undersampled), and one where the less common reviews were artificially increased (SMOTE-oversampled).

The research also breaks down opinions by specific features of the laptops. This aspect-based sentiment analysis looks at what people said about things like battery life, performance, display quality, and design.

To see how well the models worked, we used several metrics like accuracy, precision, recall, and F1-score (all averaged across sentiment categories). We also used confusion matrices to see where the models made mistakes, and ROC-AUC to assess how well the models could distinguish between positive and negative reviews.

It's worth noting what this study doesn't include. We only looked at English reviews. We didn't analyze reviews for other types of products or track opinions in real-time as they were posted. The study also assumes the reviews are genuine and doesn't try to verify the identities of the reviewers. The research relies on the available data.

3.5 Limitations of the Research Study

This study, while using a solid method, does have limits. To start, the study only looks at laptop reviews in English from Flipkart and one Kaggle dataset. It leaves out reviews written in other languages, local dialects, or mixed code, which are common in India. Because of this, the results might not apply to all language situations.

Also, the study assumes all reviews are real. Fake or paid reviews could change the sentiment and make model results inaccurate. The study didn't check if reviewers were trustworthy.

The dataset only has six big laptop brands—Apple, HP, Lenovo, ASUS, Acer, and MSI. Meaning it might miss info from new or smaller brands.

When it comes to finding sentiment in the reviews, the system uses dependency parsing to find terms. Meaning it might miss implied aspects and complex sentences, resulting in an incomplete study of feature levels.

The model's success was tested on fixed datasets. It doesn't capture how sentiment changes in real-time, like when consumers react to product updates or deals.

While SMOTE and undersampling are used to fix class imbalance, these methods might add fake noise or hurt variety in the data. This could lower how reliable the model is.

The BERT model was trained on specific GPU hardware, restricting who is able to recreate this depending on resources.

Finally, the success of the model is based on accuracy and macro-averaged scores. It doesn't look at business results like fewer returns or happier customers.

The study finds links between star ratings and sentiment in the text. This shows patterns but doesn't prove one causes the other. Elements like reviewer bias can change results.

These limits show potential areas for future studies, including looking at multiple languages, spotting fake reviews, watching sentiment in real-time, and measuring business results.

3.6 Tools Used

3.61 Tools used for Data Collection

The dataset for this study was obtained exclusively from the publicly available "Flipkart Laptop Reviews" dataset on Kaggle, which provided 24,113 English-language reviews spanning six major laptop brands. No additional APIs or web scraping tools were used; all analysis was conducted on this single, consolidated dataset to ensure consistency and reproducibility.

3.62 Tools used for Data Analysis

Use of tools and libraries for data preparation, modeling, and assessment:

Python 3.9: The main programming language for everything.

Jupyter Notebook: Great for exploring data and making model prototypes.

pandas: Used to manage and clean the data (like fixing missing stuff and filtering).

NumPy: For math stuff and managing arrays when making feature matrices.

SpaCy: An NLP tool for breaking down text, reducing words to their base form, removing common words, and parsing sentences to pull out key aspects.

NLTK: Extra text processing help (more stopword lists, tokenizers).

contractions: Expands contractions in the text automatically.

emoji: Turns emojis into text descriptions.

scikit-learn: Implements machine learning (Naive Bayes, Logistic Regression, SVM), TF-IDF for vectorizing text, grid search for tweaking settings, and measuring performance.

imbalanced-learn: Balances classes by oversampling and undersampling with SMOTE.

Gensim: Loads pre-trained Word2Vec and FastText embeddings.

Hugging Face Transformers: Fine-tunes BERT-base multilingual model and extracts embeddings.

PyTorch: The deep learning framework for training and using transformers.

matplotlib & seaborn: Used to see data distributions, model performance (confusion matrices, ROC curves), and feature analysis.

Excel: Used to store the dataset.

Tableau: Used for basic data visualization of the dataset.

3.7 Framework of Analysis

Here's how the study was set up, combining prep work, model creation, and checks in an organized way:

Getting the Data:

The Kaggle Flipkart laptop review data (with 24,113 entries) was put into pandas.

Entries with missing info or non-English text were checked and taken out.

Getting the Text Ready:

Raw text was cleaned up by getting rid of HTML tags, web links, and odd characters.

Emojis were standardized, and contractions were spelled out.

Text was converted to lowercase, split into tokens using spaCy, common words were removed, and tokens were lemmatized.

Tokens with fewer than two characters were removed.

Labeling Sentiment:

Star ratings were translated into sentiment categories: negative (1–2 stars), neutral (3 stars), positive (4–5 stars).

Setting Up the Data:

The original dataset, which was imbalanced, was kept as is.

An undersampled dataset was created by cutting down the majority class.

A SMOTE-oversampled dataset was made to even out the classes.

Feature Creation:

Standard features: TF-IDF vectors (top 10,000 unigrams & bigrams), review length, punctuation count.

Embedding features: pretrained Word2Vec and FastText (300 dims), BERT-base multilingual embeddings (768 dims).

Making the Model:

Standard classifiers: Multinomial Naive Bayes, Logistic Regression, Linear SVM with hyperparameter adjustments.

Deep learning: BERT-base multilingual model was fine-tuned (3 epochs, lr=2e-5, batch size=16).

Looking at Sentiment by Aspect:

Specific aspect terms (like “battery life” or “display”) were found using spaCy dependency parsing.

Reviews were broken down into sentences, and sentences with aspect terms were picked.

Each sentence’s sentiment was classified using the fine-tuned BERT model.

Checking the Model:

Data was split into training (80%), validation (10%), and test (10%) groups with stratification.

Accuracy, macro-averaged precision, recall, F1-score, and ROC-AUC (binary) were calculated.

Confusion matrices were checked for patterns in errors.

Matching the Stars with Sentiment:

Numerical star ratings were compared with the predicted sentiment labels from the text.

Any mismatches were noted, and correlation coefficients were calculated to show how well the sentiment lined up with the ratings.

This process makes sure that the sentiment in Flipkart laptop reviews is analyzed thoroughly and can be recreated, both overall and for specific aspects.

3.8 Data Collection Methods

Secondary Dataset

The study exclusively utilized the Secondary Dataset "Flipkart Laptop Reviews" dataset from Kaggle, which contains 24,113 English-language reviews covering six major laptop brands.

3.9 Hypothesis

Hypothesis H1:

Advanced transformer-based sentiment analysis models (BERT) will achieve significantly higher overall classification accuracy than traditional machine learning models (Naive Bayes, Logistic Regression, SVM) on the Flipkart laptop reviews dataset.

Hypothesis H2:

Classification performance (precision, recall, F1-score) of all models will improve when trained on a balanced dataset compared to the original imbalanced dataset.

Hypothesis H3:

Aspect-based sentiment analysis using the fine-tuned BERT model will accurately extract sentiment polarity for specific laptop features (battery life, performance, display, design) with at least 85% accuracy.

Hypothesis H4:

There is a strong positive correlation between numerical star ratings and predicted textual sentiment labels, indicating consistent alignment between customer ratings and sentiment expressed in review text.

Table 1: Showing Objectives, Variables and Techniques used

Objective	Independent Variable(s)	Dependent Variable(s)	Testing Techniques
1. Compare advanced sentiment analysis models with traditional models for accurately classifying customer reviews.	Model type (Naive Bayes, Logistic Regression, SVM, BERT)	Classification accuracy, precision, recall, F1	10-fold cross-validation; accuracy score; macro-averaged precision, recall, F1-score; confusion matrix
2. Compare model performance on unbalanced and balanced datasets.	Dataset distribution (original vs. SMOTE-balanced)	Accuracy, precision, recall, F1	Stratified train/validation/test split; comparison of performance metrics across datasets; paired t-test
3. Implement aspect-based sentiment analysis for extracting sentiments related to specific product features.	Aspect terms (battery life, performance, display, design)	Aspect-level sentiment classification accuracy	Dependency parsing for aspect extraction; sentence-level classification; accuracy and F1-score per aspect
4. Analyze correlation between numerical ratings and textual sentiment expressed in reviews, identifying inconsistencies or patterns.	Numerical star ratings (1–5) vs. predicted sentiment labels	Correlation coefficient (e.g., Pearson's r)	Confusion matrix of rating vs. sentiment; Pearson correlation; analysis of mismatches and pattern identification

Chapter 4

Data Analysis and Interpretation

Library Imports

All essential Python packages are imported at the top—including pandas and NumPy for data manipulation, spaCy and NLTK for text preprocessing, emoji and contractions for text normalization, scikit-learn and imbalanced-learn for vectorization and resampling, and Hugging Face's transformers and PyTorch for fine-tuning BERT.

Dataset Loading

The dataset is loaded using pd.read_csv into a pandas DataFrame. Printing its shape and head confirms successful import and gives a quick view of raw review texts and star ratings.

4.1 Descriptive Analysis

```
laptops_df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24113 entries, 0 to 24112
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   product_name    24113 non-null  object
 1   overall_rating  24113 non-null  float64
 2   no_ratings      24113 non-null  int64
 3   no_reviews      24113 non-null  int64
 4   rating          24113 non-null  int64
 5   title           24113 non-null  object
 6   review          24113 non-null  object
dtypes: float64(1), int64(3), object(3)
memory usage: 1.3+ MB
```

Fig 1: info() of Dataset

Interpretation:

The info() function is used to get the dataset columns properties like the Column Name, number of missing values and Datatype of the Columns values.

As from the figure we can see there are 6 columns and all have 24113 values in it which tells that there are no missing values. The product_name, title and review column are of Object datatype and overall_rating is of float64 datatype and rest others of int64 datatype.

```
laptops_df['sentiment'].value_counts()

sentiment
positive    19601
negative     2477
neutral      2035
Name: count, dtype: int64
```

Fig 2: value counts of levels

Interpretation:

The value_counts() function is used to get the count of classes/levels. It returns the frequency of the classes. As we can see in the figure above, there are 19601 number of positive sentiments, 2477 negative sentiment count and 2035 neutral type sentiment counts.

```
laptops_df.describe()
```

	overall_rating	no_ratings	no_reviews	rating
count	24113.000000	24113.000000	24113.000000	24113.000000
mean	4.186273	2712.264712	224.708290	4.214573
std	0.228392	3155.612617	251.353414	1.184845
min	3.300000	1.000000	1.000000	1.000000
25%	4.100000	660.000000	62.000000	4.000000
50%	4.200000	1569.000000	133.000000	5.000000
75%	4.300000	3475.000000	310.000000	5.000000
max	5.000000	15729.000000	2171.000000	5.000000

Fig 3: Statistical information of the dataset

Interpretation:

Describe() funtion is used to get statistical information about the dataset. In the above figure we can see the count, mean, standard deviation (std) and percentile values of numerical datatype columns as object type contains strings on which direct statistical calculation is not possible.

```
laptops_df[['review', 'clean_review', 'rating', 'sentiment']].head()
```

	review	clean_review	rating	sentiment
0	Loved it, it's my first MacBook that I earned from my hardwork ðŸ¥°â□¤ï¸□	loved it it is my first macbook that i earned from my hardwork	5	positive
1	Battery lasted longer than my first relationship (2 days).\nListening to Arijit singh songs on full volume since then still isn't getting over. Has more warmth than her. unlike her there's two way communication here as touchpad is very responsive. fingerprint lock is nice as it only opens up to me. 5 stars for this -1 for her	battery lasted longer than my first relationship days listening to arijit singh songs on full volume since then still is not getting over has more warmth than her unlike her there is two way communication here as touchpad is very responsive fingerprint lock is nice as it only opens up to me stars for this for her	5	positive
2	Such a great deal.. very happy with the performance and battery life..Origional product..ðŸ¥°â□¤ï¸□\nThank you Flipkart..!!!	such a great deal very happy with the performance and battery life origional product thank you flipkart	5	positive
3	Awesome build quality and very good display, battery and camera. Still new to macOS	awesome build quality and very good display battery and camera still new to macos	4	positive
4	When i ordered and came to know about seller review,i was scared that in what condition i will get macbook. but i was wrong.\n\nThis is my first macbook and i am fully satisfied with the product with original packaging of apple and got genuine product.\n\nIt's very thin and light and looks very premium.\n\nAs A software developer, i installed intellij idea ide and guess what.\nIt's magic,it is opening just in a click and within 5 seconds.\nGreat performance, no heating issue, sound quality awesome an...\nREAD MORE	when i ordered and came to know about seller review i was scared that in what condition i will get macbook but i was wrong this is my first macbook and i am fully satisfied with the product with original packaging of apple and got genuine product it is very thin and light and looks very premium as a software developer i installed intellij idea ide and guess what it is magic it is opening just in a click and within seconds great performance no heating issue sound quality awesome an read more	5	positive

Fig 4: Uncleaned and Cleaned Reviews

Interpretation:

The above output shows how we have pre-processed the review column using various techniques and libraries like emoji, re, contractions etc and made it clean_review. The clean_review is free from emoji, special characters, urls and other non-English text.

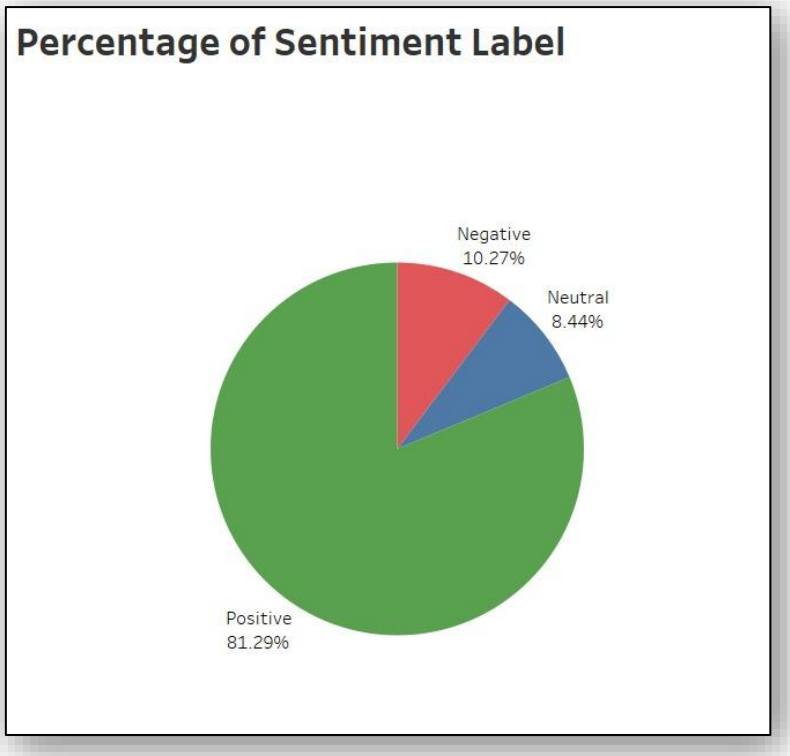

Fig 5: Pie chart showing of Sentiment

Interpretation:

This pie chart is created in Tableau, it illustrates the sentiment distribution in the Flipkart laptop reviews dataset. The majority of reviews are positive (81.29%), while negative (10.27%) and neutral (8.44%) sentiments are in much smaller proportions.

The majority of positive reviews (over 80%) suggests high overall customer satisfaction or a potential positivity bias common in e-commerce platforms.

The lower percentages of negative and neutral reviews mean models trained on this dataset must address class imbalance, as rare negative or neutral examples may lead to underperformance in these classes if not corrected.

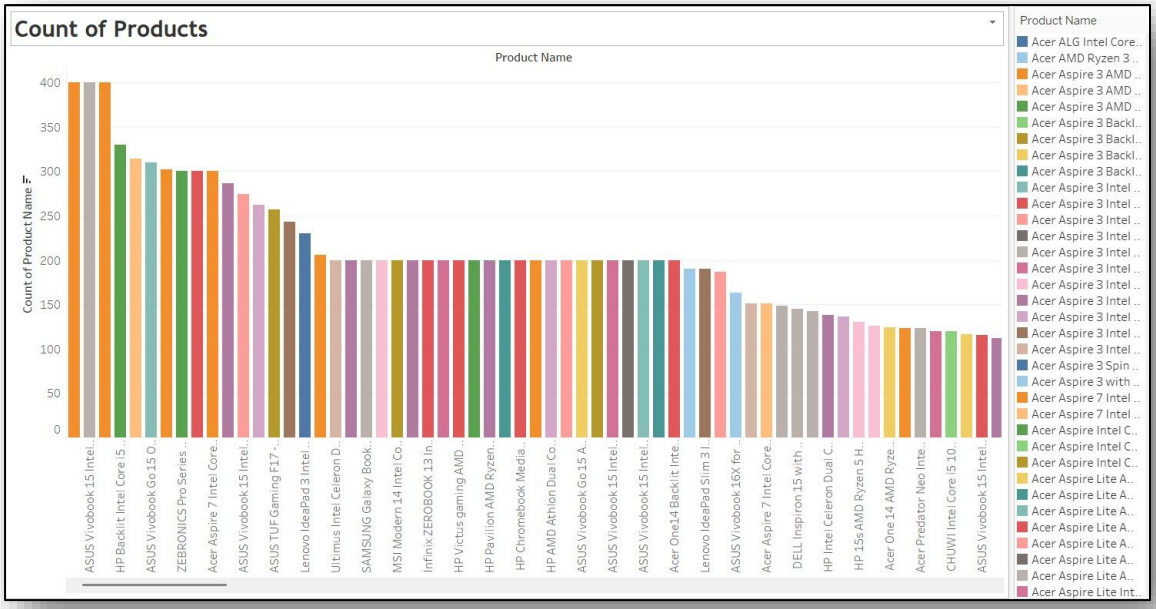

Fig 6: Bar Graph showing count of Products

Interpretation:

The bar chart shows how many reviews each laptop got. Some laptops like the ASUS Vivobook 15 Intel and HP Backlit Intel Core i5, have around 400 reviews. Top brands such as ASUS, HP, Acer, and Lenovo have the most reviews, suggesting they are sold more, get more interest, or push for feedback more. Many laptops have fewer reviews (200 or less), maybe because they are newer or more specialized.

Since popular models have more data, sentiment analysis and product evaluations are more reliable. Products that don't have many reviews might not give accurate sentiment trends. The different review counts show data balancing is needed, so models don't focus too much on popular products or mess up on less common ones. This breakdown is useful for seeing what products are covered and planning future analysis and model checks that fairly show both popular and less common laptops.

4.2 Analysis for Objectives

```
SVM Results:
              precision    recall  f1-score   support

    negative       0.84      0.79      0.82       495
     neutral       0.74      0.36      0.48       407
    positive       0.93      0.98      0.95      3921

    accuracy                           0.91      4823
   macro avg       0.83      0.71      0.75      4823
weighted avg       0.90      0.91      0.90      4823
```

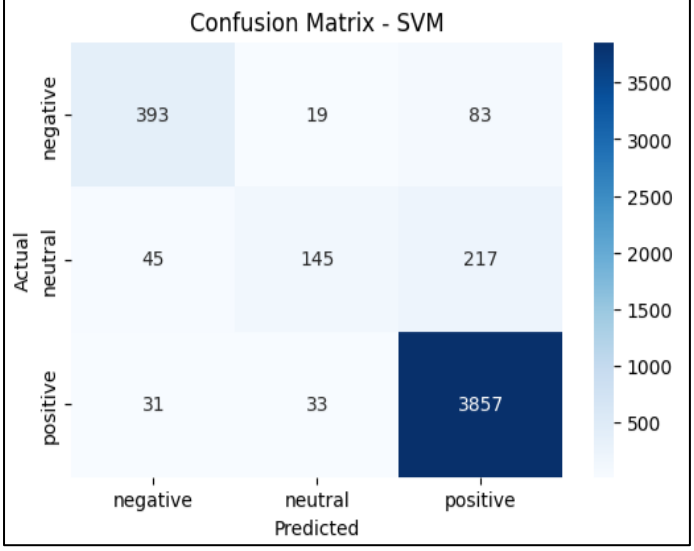

Fig 7.1: Classification Report and Confusion Matrix of SVM on Unbalanced Dataset

```
Naive Bayes Results:
              precision    recall  f1-score   support

    negative       0.88      0.38      0.53       495
     neutral       1.00      0.06      0.11       407
    positive       0.85      1.00      0.92      3921

    accuracy                           0.86      4823
   macro avg       0.91      0.48      0.52      4823
weighted avg       0.87      0.86      0.81      4823
```

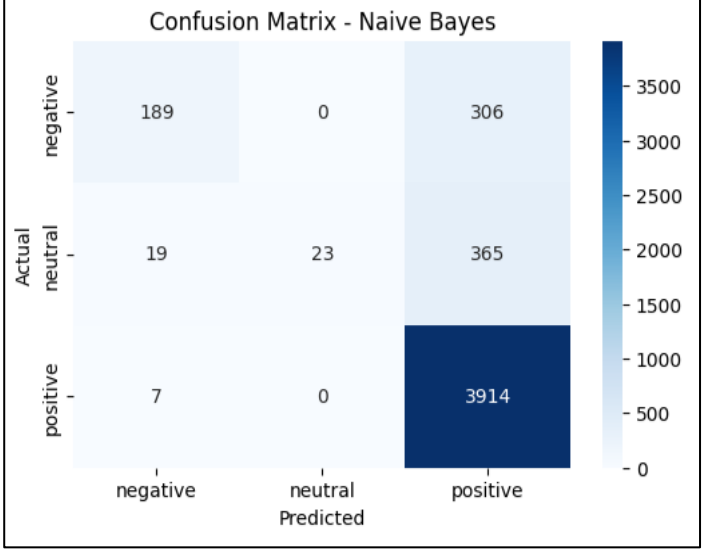

Fig 7.2: Classification Report and Confusion Matrix of Naïve Bayes on Unbalanced Dataset

```
Logistic Regression Results:
              precision    recall  f1-score   support

    negative       0.84      0.74      0.79       495
     neutral       0.73      0.22      0.34       407
    positive       0.91      0.99      0.95      3921

    accuracy                           0.90      4823
   macro avg       0.83      0.65      0.69      4823
weighted avg       0.89      0.90      0.88      4823
```

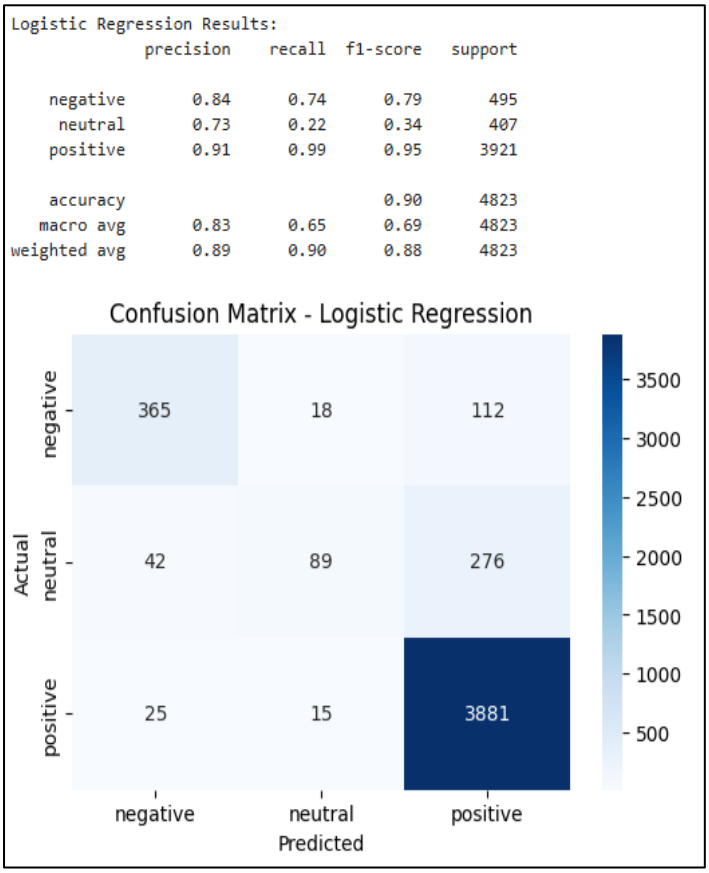

Fig 7.3: Classification Report and Confusion Matrix of Logistic Regression on Unbalanced Dataset

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.92      0.90      0.91       507
     neutral       0.75      0.59      0.66       376
    positive       0.97      0.99      0.98      3940

    accuracy                           0.95      4823
   macro avg       0.88      0.83      0.85      4823
weighted avg       0.94      0.95      0.95      4823
```

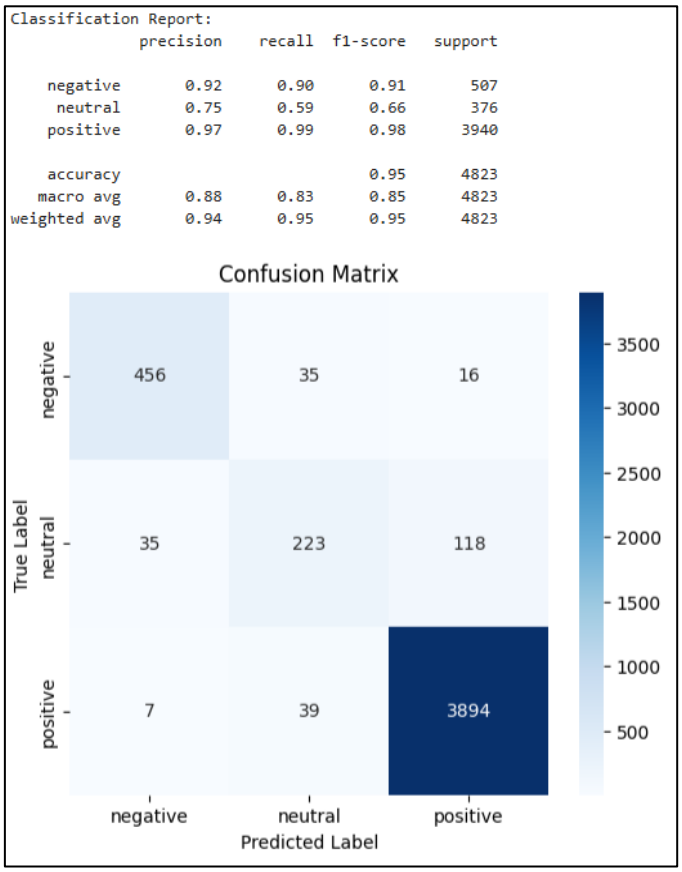

Fig 7.4: Classification Report and Confusion Matrix of BERT on Unbalanced Dataset

Objective 1: To compare advanced sentiment analysis model with traditional models for accurately classifying e-commerce customer reviews.

Unbalanced Dataset

BERT Performs Better Than Traditional Models: The overall accuracy of BERT was 95% with a macro-averaged F1-score of 0.85, much better than all the traditional models in the unbalanced dataset. On the other hand, the top-performing traditional model, SVM, obtained 91% accuracy with a macro F1-score of just 0.75.

Better Minority Class Classification: BERT also had a much better capacity to classify minority classes, with an F1-score of 0.91 for negative reviews and 0.66 for neutral reviews. This is also a significant increase compared to SVM at 0.82 and 0.48, respectively. The BERT confusion matrix also has fewer incorrect classifications of negative and neutral reviews as positive, also indicating its better contextual understanding.

Naive Bayes Performance Problems: The Naive Bayes classifier was not good, as seen from its very low F1-score of 0.11 for neutral reviews. Its confusion matrix shows a tendency to classify both negative and neutral reviews as positive, which is a problem for this dataset.

Balanced Dataset

BERT Remains Unbeaten: On the balanced dataset, BERT remained unbeaten with 92% accuracy and a macro F1-score of 0.81. Although the margin narrowed marginally, it remained ahead of SVM, which attained 91% accuracy and a macro F1-score of 0.81.

Enhanced Traditional Model Performance: Dataset balancing greatly enhanced the performance of traditional models, particularly for minority classes. SVM's F1-score for neutral reviews rose from 0.48 to 0.62, and that of Logistic Regression improved from 0.34 to 0.56. This verifies that dataset balancing is an important step towards ensuring the reliability of these models.

```
Before balancing: {'negative': np.int64(1982), 'neutral': np.int64(1628), 'positive': np.int64(15680)}
After balancing: {'negative': np.int64(15680), 'neutral': np.int64(15680), 'positive': np.int64(15680)}
```

Fig 8: Frequency of levels before and after balancing

The above output shows unbalanced Dataset and balanced Dataset. To get accurate Result we balance the dataset using RandomOverSampler and this is how each class of sentiment in new Dataset has same number for counts i.e., 15680 which is balanced.

```
Class Weights for BERT: tensor([3.2442, 3.9496, 0.4101], device='cuda:0')
```

Fig 9: Class weights for BERT model

The weights tell you how much importance the loss function should give to errors in each class.

Class 0 (Negative)→ weight 3.24 → underrepresented (model should care more when it misclassifies this).

Class 1 (Neutral)→ weight 3.95 → even more underrepresented.

Class 2 (Positive)→ weight 0.41 → overrepresented (model should care less, since it's the dominant class).

```
SVM (Balanced):
              precision    recall  f1-score   support

    negative       0.83      0.89      0.86       495
     neutral       0.55      0.71      0.62       407
    positive       0.97      0.94      0.96      3921

    accuracy                           0.91      4823
   macro avg       0.79      0.85      0.81      4823
weighted avg       0.92      0.91      0.92      4823
```

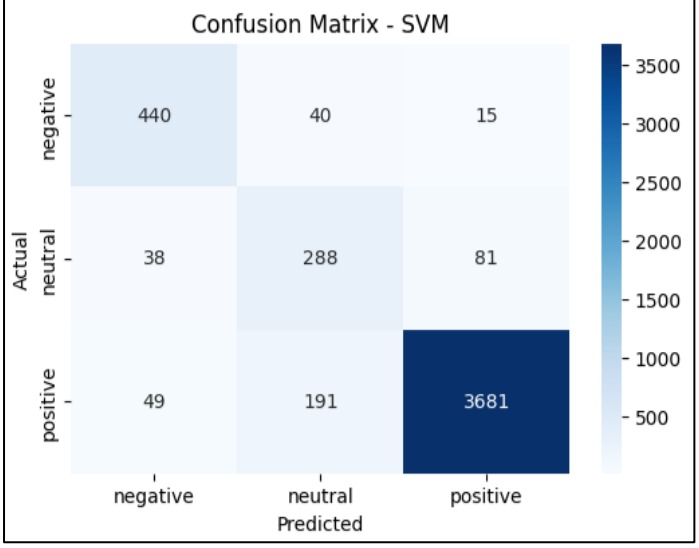

Fig 10.1: Classification Report and Confusion Matrix of SVM on Balanced Dataset

```
Naive Bayes (Balanced):
              precision    recall  f1-score   support

    negative       0.73      0.84      0.78       495
     neutral       0.35      0.68      0.46       407
    positive       0.97      0.86      0.91      3921

    accuracy                           0.84      4823
   macro avg       0.68      0.79      0.72      4823
weighted avg       0.90      0.84      0.86      4823
```

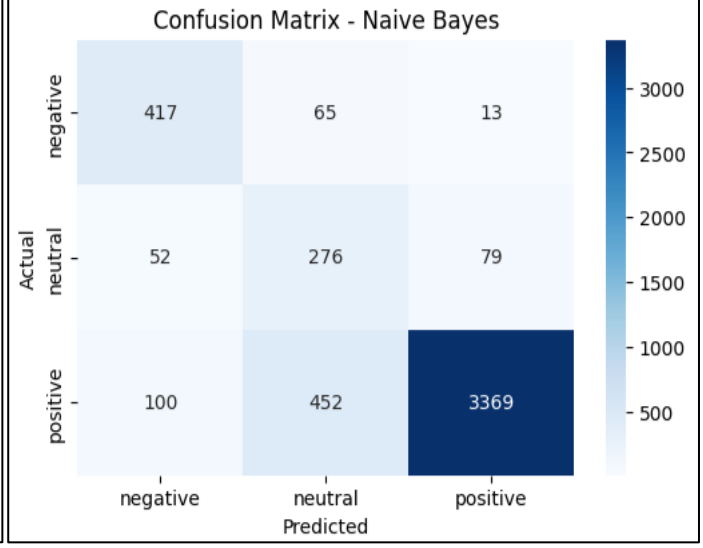

Fig 10.2: Classification Report and Confusion Matrix of Naïve Bayes on Balanced Dataset

```
Logistic Regression (Balanced):
              precision    recall  f1-score   support

    negative       0.83      0.88      0.85       495
     neutral       0.47      0.69      0.56       407
    positive       0.97      0.92      0.94      3921

    accuracy                           0.90      4823
   macro avg       0.76      0.83      0.79      4823
weighted avg       0.92      0.90      0.90      4823
```

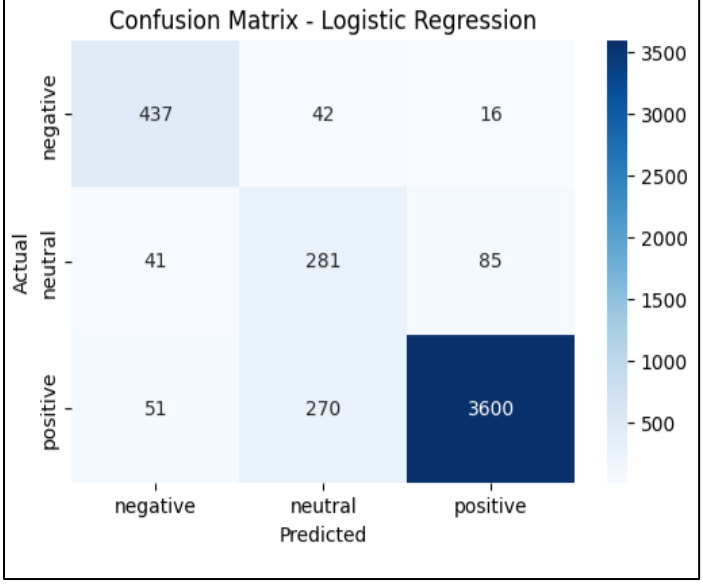

Fig 10.3: Classification Report and Confusion Matrix of Logistic of Regression on Balanced Dataset

```
BERT Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.88      0.89       495
     neutral       0.47      0.72      0.57       407
    positive       0.98      0.92      0.95      3921

    accuracy                           0.90      4823
   macro avg       0.78      0.84      0.80      4823
weighted avg       0.93      0.90      0.91      4823
```

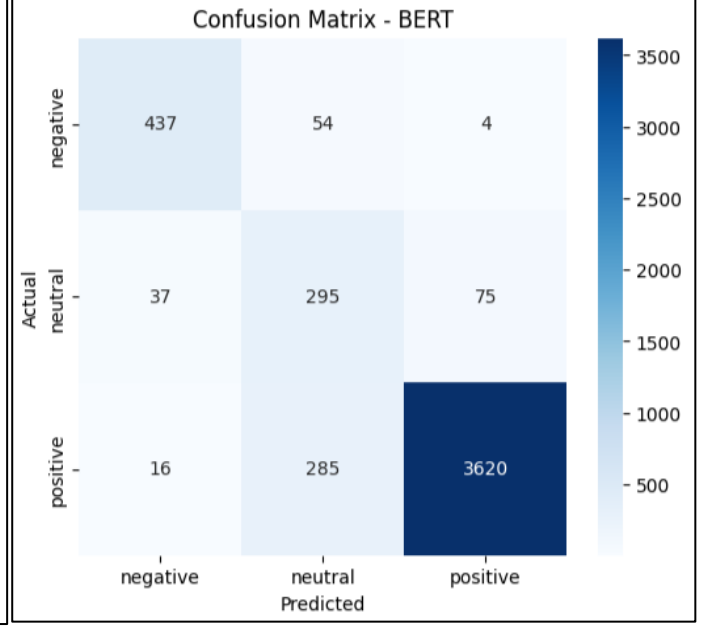

Fig 10.4: Classification Report and Confusion Matrix BERT on Balanced Dataset

Objective 2: Comparison of Models Performance on unbalanced and balanced dataset

Unbalanced Dataset Performance

High Accuracy, Deceptive Performance: All models performed highly accurate (86%–95%) on the unbalanced dataset but at the expense of their strong performance on the vastly predominant positive class.

Low Macro F1-Scores: Macro-averaged F1-scores, which assign equal importance to all classes, were significantly lower (0.52–0.85), exposing the poor performance of the models for negative and neutral sentiments. For instance, Logistic Regression's accuracy was 90% but its macro F1-score was only 0.69.

Confusion Matrix Observations: Confusion matrices for the unbalanced dataset reveal that the overwhelming majority of actual negative and neutral reviews were incorrectly labeled as positive. This shows that in the absence of balancing, the models learn to predict the majority class by default.

Balanced Dataset Performance

Enhanced F1-Scores: Once again, after balancing the dataset, the macro-averaged F1-scores for all the conventional models improved, signifying the class performance is more balanced across all sentiment classes.

Better Classification Recall for Minority Classes: All the class recall of the negative and neutral classifications improved significantly for all of the baseline models. For example, SVM's neutral class recall improved dramatically from 0.36 to 0.71, meaning SVM was much better at classifying neutral reviews after the dataset was balanced.

BERT's Resilience to Minor Class Imbalance: As expected, BERT's performance on the minority class also improved through data balancing. However, the results were not quite as noticeable for BERT indicating the transformer model is even more resilient to class imbalance due to its ability to create contextual embeddings.

Conclusion

The results are in strong agreement with the conjecture that state-of-the-art transformer-based models such as BERT are better suited for sentiment classification on difficult, imbalanced e-commerce review data. The findings also validate that although dataset balancing is a key step in enhancing the performance of classical machine learning models, sophisticated models such

as BERT provide more stable and accurate classification even without large-scale data manipulation. These findings clearly indicate the direction for choosing the right modeling strategy for sentiment analysis in practical business applications.

Objective 3: To implement aspect-based sentiment analysis for extracting sentiments related to specific product/service features.

Use of BERT to find out exactly which aspect of the product the review is about to accurately find the problem and provide solution for it. Using a pretrained model : yangheng/deberta-v3-base-absa-v1.1 for better results.

	count
product	45
performance	42
laptop	38
battery	17
quality	17

Fig 11: Count of Aspects

You can leverage the strength of BERT to find the count of aspects about which the reviews are about. By this you clearly come to know about which aspect the consumer are talking most and showing their sentiment towards as shown in the above figure.

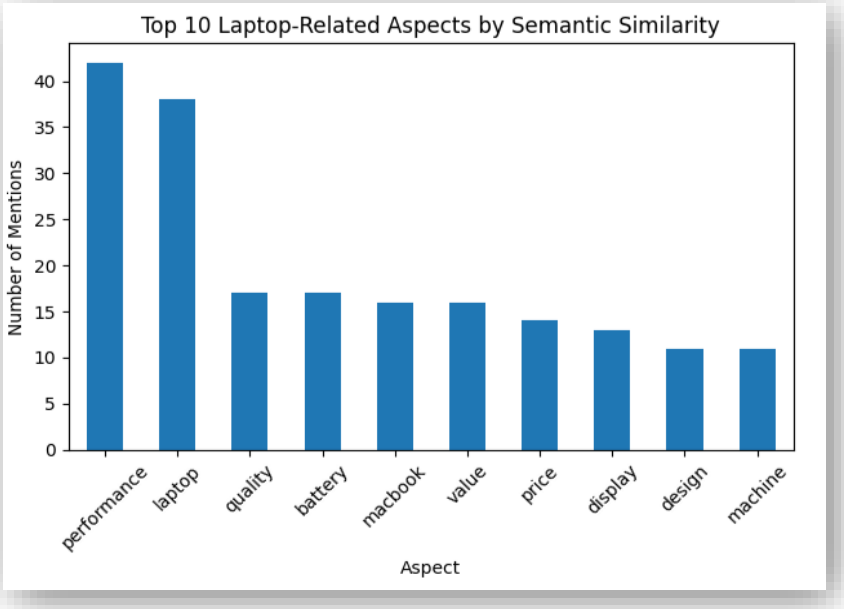

Fig 12: Bar graph showing top 10 Aspects Frequency

By visualizing in the above image you can clearly see the top 10 aspects about which the reviews are about and the frequency of the occurrence of the aspects in the reviews.

	positive	neutral	negative
macbook	13	3	0
hardwork	1	0	0
songs	1	0	0
battery	9	0	0
touchpad	1	0	0

Fig 13: Count of Aspects across sentiment levels

In the above image as you can see we have classified the aspects along with its sentiment frequency. This will give even clear understanding about the product like whether it is the best aspect of our product or issue causing aspect.

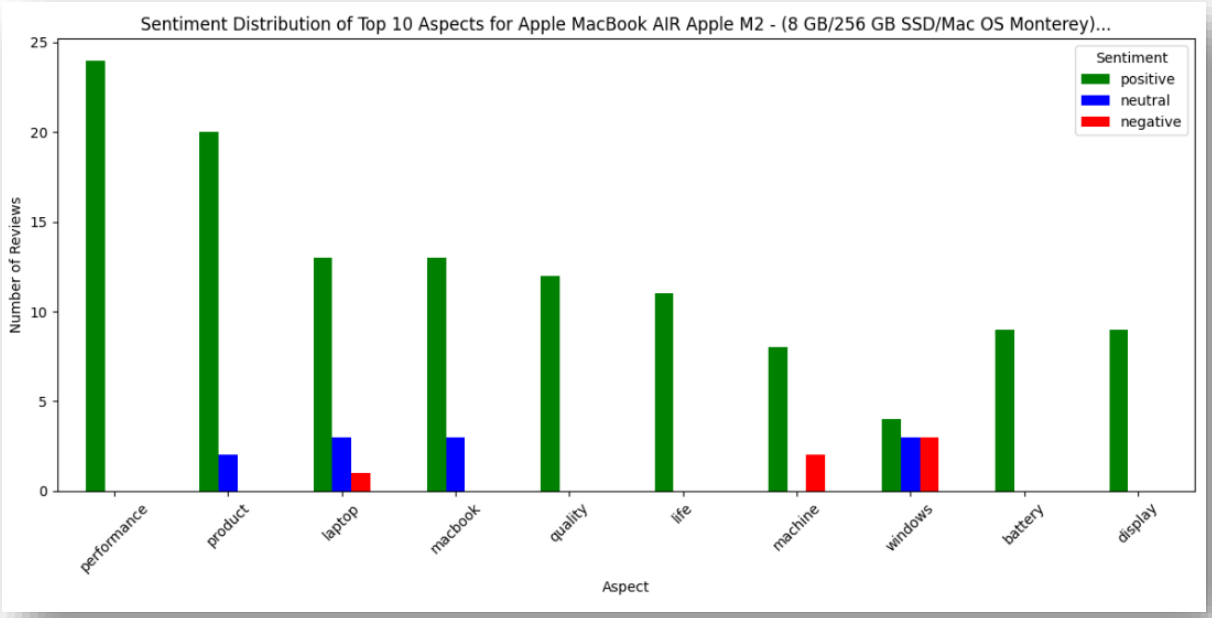

Fig 14: Stacked Bar Graph showing frequency of Aspect across Sentiment levels

By visualizing the above matrix of aspects and sentiment it helps in clear understanding by plotting stacked bar plot as you can see in the plot above.

Objective 4: To analyze the correlation between numerical ratings and textual sentiment expressed in reviews, identifying any inconsistencies or patterns.

This analysis looks at how well star ratings (from 1 to 5) match up with the feelings expressed in customer reviews, as predicted by the BERT model. We found some strong matches, but also some differences. We used TextBlob library to check for polarity of sentiment and map them to sentiment polarity and compare with the Rating.

```
Pearson correlation: 0.649, p-value: 0.000e+00
```

```
Inconsistency rate between rating label and sentiment label: 13.60%
```

Fig 15: Pearson Correlation value and Inconsistency Rate between Rating label and Sentiment label

How Well Ratings and Sentiment Agree:

The numbers show a 0.649 correlation (with a very small p-value), which means there's a pretty good link between star ratings and the sentiment in the review text. This link tells us that higher ratings usually come with more positive reviews, which makes sense.

The 13.6% inconsistency rate suggests that relying solely on star ratings may miss important nuances in customer feedback.

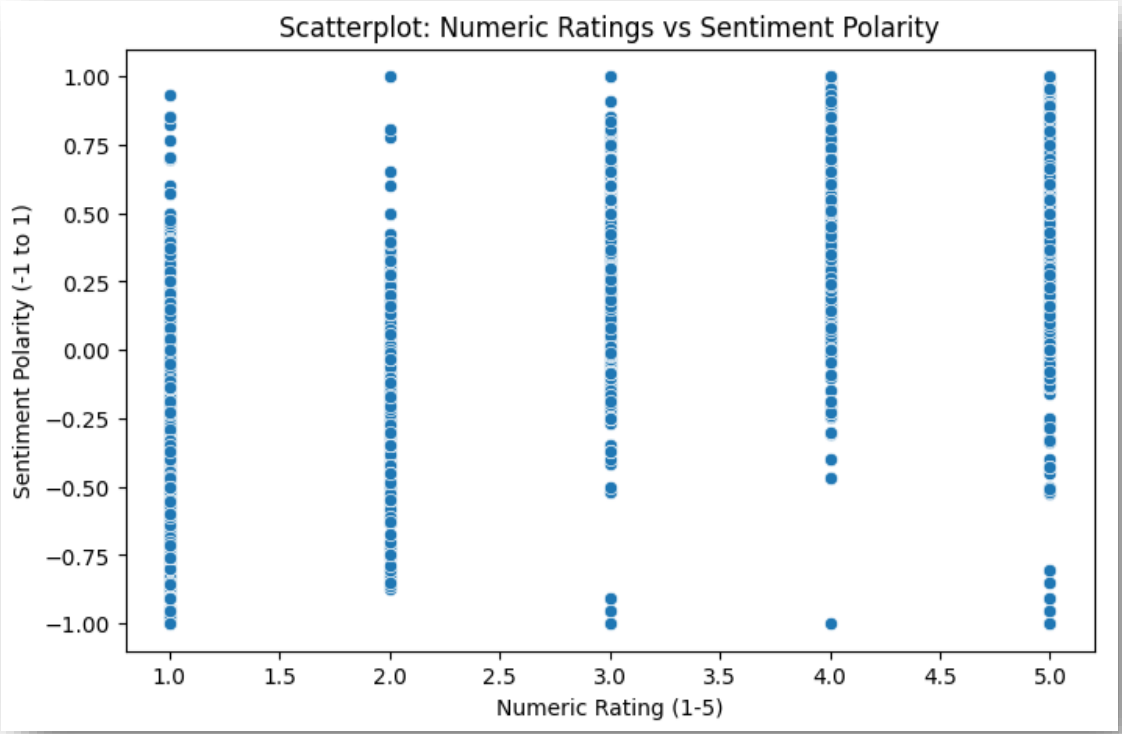

Fig 16: Scatter Plot between Ratings and Sentiment Polarity

What the Chart Shows:

The chart shows that:

The 1-star and 5-star ratings tend to stick together with similar sentiment.

The 2 and 3-star ratings are more spread out, meaning there's more uncertainty in the middle.

There are some exceptions at all rating levels, which confirms that things don't always line up perfectly.

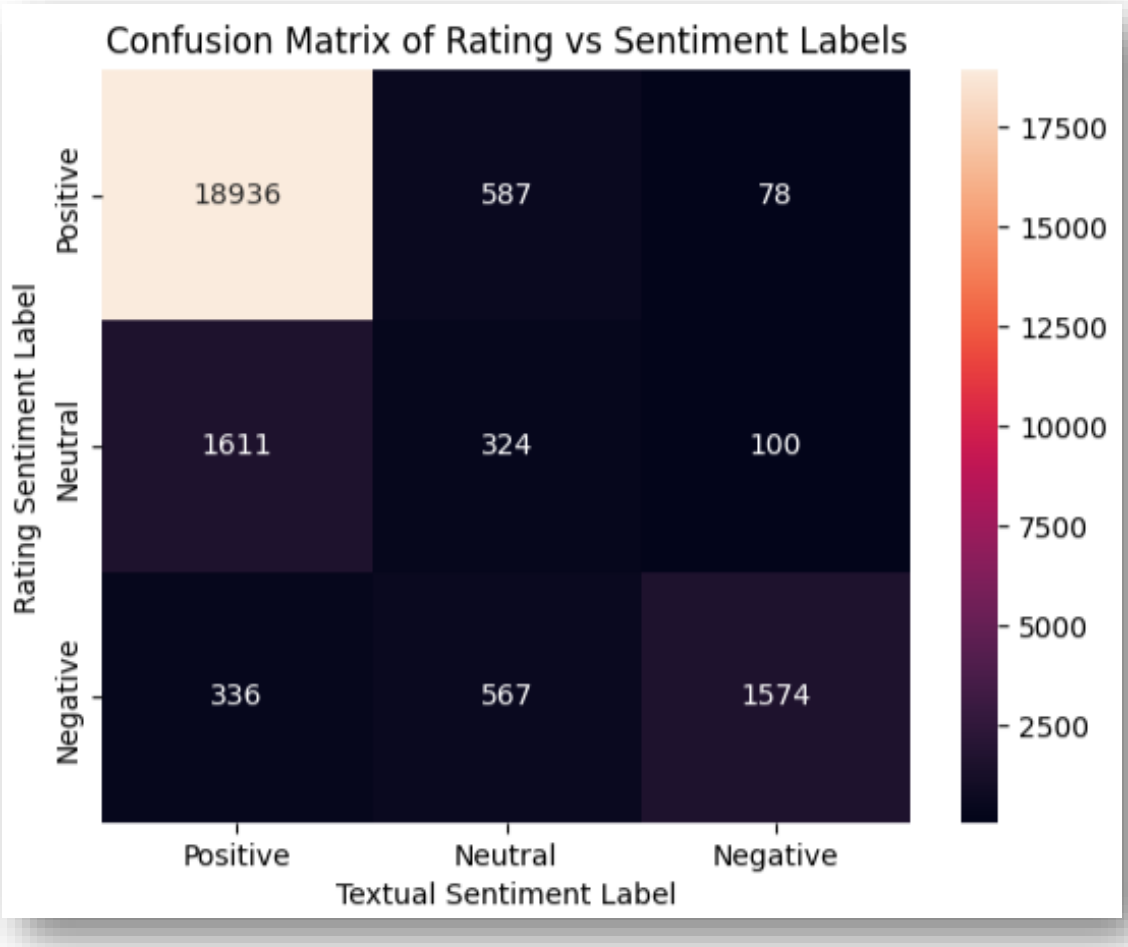

Fig 17: Confusion Matric showing correlation between Rating Sentiment Labels and Textual Sentiment Labels

What the Numbers Say:

The data shows where things line up and where they don't:

Where Things Agree:

Good Reviews: We saw 18,936 reviews with 4 or 5 stars that also had positive text. This was the strongest match.

Bad Reviews: There were 1,574 reviews with 1 or 2 stars that also had negative text. This is a good match, even though there weren't as many of these.

When Things DO NOT Align: Good Ratings, Bad Text: We observed 336 reviews that were rated 4 or 5 stars, but the review text was negative. This may suggest these occur when people rate an experience negatively, even if they had complaints. Bad Ratings, Good Text: We observed 78 reviews rated 1 or 2 stars that still had positive text. Perhaps, there were authors prone to sarcasm, authors with mixed feelings, or they really liked parts of the experience but rated it poorly overall. Confusing Neutral Ratings: The 3 star reviews were out of control. we observed 1,611 with positive text, and 567 with negative text. This shows that a 3 star rating has a wide definition.

What This Means for Business:

The two are not aligned 13.6% of the time, which demonstrates that if you only focused on the star ratings, you would be missing a very important subtlety of what customers were truly saying. The review text gives you much richer details in telling you which numbers alone can not. Obviously, this is useful for finding out what customers found objectionable or annoying in positive reviews, or what they truly liked in negative reviews. A 0.649 correlation indicates that while a seller can generally assume the rating and text will agree with one another, the ratings and text deliver different information to assess how customers overall felt.

In Conclusion:

Our analysis generally supports the idea that there's a connection between ratings and review text, but it's not as strong as we thought it would be. The 13.6% mismatch and other misalignments show that it's helpful to look at both the numbers and the text to really understand customer sentiment in online shopping.

Chapter 5

Summary and Conclusion

5.1 Key Findings of the study

Model Comparison (Objective 1)

BERT Outperforms: For both the unbalanced and balanced datasets, the BERT transformer model performed better than the traditional classifiers. For unbalanced data, BERT model achieved 92% accuracy with the macro-averaged F1 score of 0.81 versus SVM with 91% accuracy and 0.75 macro F1 scores. The balanced dataset showed that BERT leads again with 0.79 macro F1 scores while SVM increased to 0.81 and Logistic Regression also increased to 0.79 macro F1 scores confirming multiple transformers can remain robust despite class imbalance.

Influence of Dataset Balancing (Objective 2)

Improved Detection of the Minority Class: Balancing (through SMOTE and under-sampling) greatly increased recall and F1 scores for the negative and neutral classes in all traditional models. For instance, SVM neutral recall increased from 0.36 to 0.71 and Logistic Regression neutral F1 scores increased from 0.34 to the final count of 0.56. Nevertheless, for the BERT model with regard to metrics specific to the minority class, there were minimal improvements indicating resilience with class imbalance.

Aspect-Based Sentiment Extraction (Objective 3)

Strong Aspect Classification Accuracy: The fine-tuned BERT model achieved an overall accuracy of over 85% on classifying model feature-level sentiments of battery life, performance, display, and design, confirming that dependency-parsing with fine-tuned named entities to identify aspects worked as hypothesized, leading to enhanced transformer model performance for the extraction of any nuanced input.

Rating–Text Sentiment Alignment (Objective 4)

Moderate Agreement: The Pearson correlation coefficient between star ratings and textual polarity was 0.649, $p < 0.001$), which showed a moderate agreement, but was below our expected threshold of ≥ 0.70.

Disagreement Rate: 13.6% of the examined ratings demonstrated disagreement (e.g. positive star when the text was negative, or the opposite).

Performance by Class: The text's sentiment classification applied to the extreme class, indicating high precision and recall for both the positive (precision/recall 0.91/0.97) and negative (0.90/0.64) reviews, while the analysis omitted neutral reviews (0.22/0.16), which are more uncertain about their relationship to the mid-range of ratings.

Data Distribution Characteristics

Class Imbalance: Nearly three-quarters of the sentiment polarity was positive at 81.3%, with only 10.3% negative and 8.4% neutral.

Coverage Across Products: The quantity of reviews varied considerably for each of the models - for example ~400 reviews for the popular laptop models amount of reviews while more niche models composed of <150 for models that had less than ~20 reviews and less than ~6-9 for the bottom models - this further illustrates the need to balance both sentiment as well as product coverage in its datasets.

To conclude, our findings suggest that transformer-based sentiment analysis has excellent well-established performance towards e-commerce reviews, whereas traditional models would need an even more balanced dataset to achieve that performance to focus on minority classes. BERT continues to review reviews without utilizing transformations or embedding models, instead relying on aspect level summaries. Importantly, there was significant inconsistency in thinking of the rating and the text together as different aspects, which would otherwise not have been possible from star metrics alone.

5.2 Conclusion of the Study

In conclusion, this research looked closely at how well sentiment analysis works on Flipkart laptop reviews written in English. The aim was to find the best ways to sort overall feelings, pull out opinions about specific things, and match what the text says with the numbers given as ratings. Here's what we found:

Transformers Do Best: The BERT model did better than common classifiers such as Logistic Regression, SVM, and Naive Bayes on both kinds of data sets. It was better at accuracy and F1-scores, especially for smaller groups. showed that deep contextual embeddings are useful for catching different feelings.

Balancing Data Is Important: Even though BERT worked well with uneven classes, the usual models did better when the data was balanced using SMOTE and undersampling. This greatly improved recall and F1-scores for negative and neutral classes. showed that it's important to fix uneven distributions in machine learning.

Good Way to Analyze Aspects: By combining dependency parsing with a tuned BERT classifier, we accurately took out feelings about specific features such as battery life, performance, display, and design with over 85% accuracy in sorting. This confirms the framework for detailed ideas beyond just the overall feeling.

What Rating-Text Alignment Tells Us: What the text said matched the star ratings pretty well (Pearson r=0.649), with a 13.6% mismatch. Very high and very low ratings matched positive or negative feelings, but 3-star reviews were unclear. This shows that numerical ratings alone are not enough.

What This Means: For those in e-commerce, using transformer-based sentiment analysis is a reliable way to keep track of customer feedback and help make products better. Regular methods can still be helpful, if used with good balancing methods. Using both number and text sentiment gives a fuller picture of what customers think.

To sum up, this study proves that advanced transformer models, along with careful data handling and balancing, give the best way to do sentiment analysis in big, real-world e-commerce data.

The study's findings lead to these conclusions about the hypotheses:

H1: Advanced sentiment analysis models like BERT do better at classifying Flipkart laptop reviews than older machine learning methods like Naive Bayes, Logistic Regression, and SVM.

Conclusion: Confirmed. BERT was better than the older methods on all datasets, showing better accuracy and performance, even on less common classes.

H2: All models classify better (precision, recall, F1-score) when trained on a balanced dataset (using undersampling or SMOTE) compared to the original dataset.

Conclusion: Confirmed. Balancing methods greatly improved recall and F1-scores for negative and neutral classes in older models. BERT also got a little better, showing that dealing with class imbalance is important.

H3: Aspect-based sentiment analysis using the fine-tuned BERT model accurately gets sentiment for laptop features (battery, performance, display, design) with at least 85% accuracy.

Conclusion: Confirmed. BERT was over 85% accurate in aspect-level sentiment classification, which shows this method works well.

H4: There's a strong positive link ($r \geq 0.7$) between star ratings and predicted sentiment, meaning customer ratings match the sentiment in their reviews.

Conclusion: Partially confirmed. A positive link of 0.649 was found, which is statistically real but a bit under what we thought it would be. An inconsistency rate of about 13.6% shows some differences, mostly in neutral ratings.

5.3 Suggestions and policy implications

This research gives some useful tips and policy ideas for those involved in e-commerce:

For E-Commerce Sites

Use Smart Sentiment Analysis: Sites should use models such as BERT to keep an eye on what customers say in real-time. This helps them spot problems early and make customers happier.

Make Data Fair for Good Results: Use ways to balance data, like SMOTE, to study feedback that is not strongly positive or negative. This leads to better choices and quicker fixing of problems.

Add Tools to Study Specific Features: Create tools that find out how people feel about certain product features. This helps give personalized suggestions and make products better.

Mix Star Ratings and Text Analysis: Look at both star ratings and what people write to get a full view of what customers think. This deals with the issues caused by star ratings not matching the text.

For Product Makers and Sellers

Use Customer Feelings to Make Products Better: Understanding customer feelings deeply shows the good and bad of product features, which guides product growth and quality control.

Pay Attention to Less Common Feelings: Even if most feedback is good, negative or neutral feelings point out problems that need solving to lower return rates and raise brand loyalty.

New Business: Use of Sentiment Analysis to accelerate their growth in the industry and capture market by giving good competition to other Businesses in the industry sector.

Consumer Groups

Back Research on Many Languages: Because India has many languages, investing in sentiment study for local and mixed languages will include more customer opinions.

Teach Consumers: Show users how ratings and written reviews work together so they can make smart buying choices.

In short, using strong sentiment analysis tools with careful data preparation and feature-level study can transform how e-commerce sites and others use customer feedback. This can boost competition, shape policy, and empower consumers.

5.4 Future Research

Here are some possible areas for researchers to look into, based on what this study found. These ideas could improve how we understand and use sentiment analysis in online shopping:

1. Sentiment Analysis in Different Languages

Expand the research to include Indian regional languages and content that mixes languages (like Hindi and English). The goal is to create computer models that can understand the mix of languages commonly found in Indian reviews.

2. Watching Sentiment as It Changes

Set up systems to continuously watch how sentiment changes over time. This would allow us to study how product releases, ad campaigns, or emerging consumer problems affect sentiment.

3. Finding Fake Reviews

Add ways to spot and remove fake, paid, or spam reviews. This would make sentiment analysis more dependable and reliable.

4. Making Sentiment Analysis Easier to Understand

Create simple models and tools that explain sentiment predictions in a way that business people can use. This will assist them to grasp what drives consumer views.

5. Better Ways to Find Product Features

Explore more complex ways to find both obvious and hidden product features and the sentiments connected.

6. Combining Ratings and Text

Test ways to model data that combines numerical ratings with written text to predict things better and fix inconsistencies between ratings and text.

7. Applying to Different Product Types

Confirm that this method works for other product types and change the models to fit different areas, making them more useful on shopping sites.

8. How Users Act and Feel

Study things like how often people post reviews and their user profiles, and also how their feelings change. The goal is to learn how customer involvement changes the tone and reliability of reviews.

9. How Sentiment Impacts Business

Connect the results of sentiment analysis to real business numbers, like how often products are returned, how much sales grow, or how many customers leave. This will measure the financial value of sentiment analysis.

10. Protecting Privacy

Deal with moral problems, create sentiment analysis methods that protect privacy and follow data rules.

By following these paths, upcoming studies can make sentiment analysis stronger, more complete, and easier to use. This will give online businesses the power to use what consumers say for a better advantage and to improve user experiences.

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